

EMPLOYEE TABLE:

Emp ID	FNAME	LNAME	Salary	Dept ID
1	Arun	Kumar	50000	2
2	Riya	Patil	30000	3
3	Alina	Das	45000	2
4	Mia	Thomas	55000	1
5	Priya	Reddy	25000	3

DEPARTMENT TABLE:

Dept ID	Dept Name	Location
1	HR	Bengaluru
2	Sales	Mumbai
3	IT	Hyderabad

Relational Algebra Expressions

1. Employees in department ID 3

$\pi_{FNAME, LNAME}(\sigma_{DeptID=3}(EMPLOYEE))$

FNAME	LNAME
Riya	Patil
Priya	Reddy

2. Employees with salary > 30000

$\pi_{FNAME, Salary}(\sigma_{Salary>30000}(EMPLOYEE))$

FNAME	Salary
Arun	50000
Alina	45000
Mia	55000

3. Department name located in Bengaluru

$\pi_{\text{DeptName}}(\sigma_{\text{Location}=\text{"Bengaluru"}}(\text{DEPARTMENT}))$

Dept Name
HR

4. Employees in dept ID:2 and with salary>40000

$\pi_{\text{FNAME}, \text{Salary}}(\sigma_{\text{DeptID}=2 \text{ AND } \text{Salary}>40000}(\text{EMPLOYEE}))$

FNAME	Salary
Arun	50000
Alina	45000

5. Employees with salary<30000

$\pi_{\text{EmpID}, \text{FNAME}}(\sigma_{\text{Salary}<30000}(\text{EMPLOYEE}))$

Emp ID	FNAME
5	Priya